Assignment 01: Unit-2 Control Flow Statements (V2)

Course: C Programming & Computer Concepts

Topic: Control Flow Statements (Loops, Decision Making & Pattern Printing)

Section A: Basic Control Flow & Digit Manipulation

Question 1: Digit Reversal

Write a complete C program to reverse the digits of an integer entered by the user and store the reversed value in a separate variable before outputting it.

- **Input:** An integer *n* (e.g., 1234)
- **Output:** Reversed integer (e.g., 4321)
- **Requirements:** Extract digits using the modulo operator (% 10) and rebuild the number using iterative multiplication (reverse = reverse * 10 + digit).

Question 2: Armstrong Number Check

Write a C program to check whether a given positive integer is an Armstrong number or not. An Armstrong number of 3 digits is an integer such that the sum of the cubes of its digits is equal to the number itself.

- **Input:** An integer *n* (e.g., 153)
- **Output:** 153 is an Armstrong number or Not an Armstrong number
- **Formula:** $1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$

Question 3: Print First N Natural Numbers

Write a C program using a do-while loop to read an integer *n* from the user and print all natural numbers from 1 up to *n*.

- **Input:** An integer *n* (e.g., 5)
- **Output:** 1 2 3 4 5

Question 4: Multiplication Table Generator

Write a C program using a loop construct to print the multiplication table of a user-specified

integer *n* up to 10 multiples.

- **Input:** *n* = 5
- **Output:** 5 x 1 = 5, 5 x 2 = 10, ..., 5 x 10 = 50

Question 5: Sum and Product of Digits

Write a C program to calculate and display both the sum and the product of the individual digits of a given positive integer.

- **Input:** *n* = 1234
- **Output:** Sum = 10 (1+2+3+4), Product = 24 (1 × 2 × 3 × 4)

Question 6: Separate Sum of Odd and Even Numbers

Write a C program to calculate the sum of all odd numbers and the sum of all even numbers independently in the range from 1 to *n*.

- **Input:** *n* = 6
- **Output:** Sum of Odd = 9 (1+3+5), Sum of Even = 12 (2+4+6)

Section B: Series Evaluation Problems

Question 7: Fibonacci Series

Write a C program using a for loop to generate and display the first *n* terms of the Fibonacci sequence starting from 0 and 1.

Sequence: 0, 1, 1, 2, 3, 5, 8, 13, ...

Question 8: Harmonic Series Summation

Write a C program to calculate the sum of the following harmonic series up to *n* terms:

S = 1 + 1/2 + 1/3 + 1/4 + ... + 1/n

Question 9: Sum of Factorials Series

Write a C program to compute the sum of the factorials of numbers from 1 to *n*:

S = 1! + 2! + 3! + 4! + ... + n!

Question 10: Sum of Inverse Factorials Series

Write a C program to evaluate the sum of inverse factorials up to *n* terms:

S = 1/1! + 1/2! + 1/3! + ... + 1/n!

Section C: Nested Loop Pattern Printing

Overview of Pattern Problems

The following table summarizes the structural patterns required to be implemented using nested for loops in C:

Q#	Pattern Description	Sample Output Target
11	Dynamic Square Star Grid ($n \times n$)	`* * *` `* * *` `* * *`
12	Fixed 4x4 Star Grid	`* * * *` `* * * *` `* * * *` `* * * *`

13	Inverted Right-Angled Star Pyramid	`* * * * *`
		`  * * * *`
		`    * * *`
		`      * *`
		`        *`

14	Continuous Character Matrix	`A B C`
		`D E F`
		`G H I`

15	Repeating Character Row Matrix	`A B C D` `A B C D` `A B C D` `A B C D`

Q#	Pattern Description	Sample Output Target

16	Right-Angled Star Pyramid	`*`
		`* *`
		`* * *`
		`* * * *`

17	Semi-Pyramid of Incremental Column Numbers	`1`
		`1 2`
		`1 2 3`
		`1 2 3 4`

18	Continuous Character Right-Angled Triangle	`A`
		`B C`
		`D E F`

19	Repeating Letter per Line Triangle	`A`
		`B B`
		`C C C`

20	Fixed Column Index Number Matrix	`1 2 3 4` `1 2 3 4` `1 2 3 4` `1 2 3 4`

21	Floyd's Triangle (Sequential Numbers)	`1`
		`2 3`
		`4 5 6`
		`7 8 9 10`
		`11 12 13 14 15`

Q#	Pattern Description	Sample Output Target

22	Descending Row Index Triangle	`4 4 4 4`
		`3 3 3`
		`2 2`
		`1`

23	Hollow Star Border Box (*r* × *c*)	`* * * *`
		`*  *`
		`*  *`
		`* * * *`

24	Right-Aligned Star Pyramid	`  *`
		`  * *`
		`  * * *`
		`* * * *`

25	Escalating Character Triangle	`A`
		`A B`
		`A B C`
		`A B C D`

26	Constant Character Column Grid	`A A A`
		`B B B`
		`C C C`

27	Inverted Right-Angled Number Triangle	`1 2 3 4`
		`1 2 3`
		`1 2`
		`1`

Q#	Pattern Description	Sample Output Target

28	Inverted Right-Angled Star Triangle	`* * * *`
		`* * *`
		`* *`
		`*`